Hybrid Dev Workflow for StratumRO-QGIS (additive plan)

Goal

Add a **development and compute plane** on top of your architecture v2, without changing it. The diagram, `L1` – `L6`, the 6 golden rules, `canon/` and the existing code stay exactly as they are. Everything below is **new files or append-only edits**.

The workflow:

Tool	Role
Antigravity IDE / 2.0 (`agy`)	Home base. Gemini Flash handles routine bulk work, and `/goal`, `/plan` and `/browser` handle long, planned or research tasks.
Colab CLI	Heavy GPU jobs: L2a training, L2b SAM 2.1 batch refinement, L5 VLM experiments, retraining from the feedback loop, and optional bulk tiling for L1.
Claude Pro	Architecture, hard PyQGIS/geometry/training bugs, code review and the L3/L5/MCP logic.
Copilot (Student)	Inline completions, boilerplate, small edits and tests.

What I found in the repo

- Already in place:
  - is the shared rule file.
  - `CLAUDE.md` and `GEMINI.md` are one-paragraph pointers to it.
  - `.agents/` has 18 agents, 14 skills and 4 rules.
  - `mcp_config.json` has 8 servers.
  - `stratumro-brain/` holds the shared memory, with the rule that agents write to `inbox/` only.
  - CI runs 4 jobs on Ubuntu and Windows.
- Missing:

    - `.github/copilot-instructions.md`.
    - Any Colab job scaffolding. is hard-coded to `C:\Users\lefpa\Desktop\...`.
    - An Antigravity workspace MCP file at `.agents/mcp_config.json`. Only the root `mcp_config.json` exists.
    - SAM **2.1** weights. `models/sam2/` holds `sam2_hiera_tiny.pt`, which is the 2.0 checkpoint.
- Hardware: the local GPU is an **RTX 4050 with 6 GB**. The "Workstation 24 GB" in the diagram is not this machine. Colab fills that gap for training.

Research findings that change the plan

> **WARNING**
>
> These came from the web research. Items marked *unverified* need a quick test during execution.

Finding	Consequence	Verdict
`google-colab-cli` (v0.7.5, 2026-10-20) is **Linux/macOS only, Python ≥ 3.12**.	Run it in **WSL2** and call it from PowerShell through `wsl`.	Adopt
Colab VS Code extension (`google.colab`) works for notebooks only.	Use it for interactive exploration. Use the CLI for batch jobs. *Not verified to install in the Antigravity IDE.*	Adopt for notebooks
`colab-mcp` needs a Colab tab open in a browser and is not headless.	Skip for now.	Optional
Colab VMs are deleted when idle and have a maximum lifetime of about 12 hours.	Checkpoint to Drive every N epochs and make jobs resumable.	Required
Claude models inside Antigravity are documented as an **Ultra-tier** feature.	Plan to use Claude through **Claude Code or the Claude app** in the same repo. If you can already pick Claude inside Antigravity, I'll adjust.	Check with you
Copilot Student is **Auto model selection only**. The credit allowance is not published.	Use it for completions and small edits, not for heavy agent work.	Adopt, limited
Claude Pro has a 5-hour session cap and a weekly cap, shared with claude.ai.	Spend it on design, debugging and review. Keep sessions short.	Adopt
Antigravity reads `AGENTS.md` and `GEMINI.md` natively. It was **not confirmed** to read `CLAUDE.md` . `.agents/rules/*.md` files without valid frontmatter are **silently dropped**.	Keep `AGENTS.md` as the single source. Check the 4 existing rules for frontmatter.	Verify
`smp` 0.5.0 supports `in_channels=4` for UNet++ and DeepLabV3+.	Matches L2a. Use a spatial split, photometric augmentation on RGB only, and never jitter nDSM.	Adopt
`sam2.1_hiera_tiny` has 38.9M parameters. There is no official VRAM figure.	Measure `max_memory_allocated()` on the laptop and on Colab.	Adopt, measure
QGIS 3.40 Docker images stopped updating in 2026-01. `3.44` (LTR) and `4.0` exist.	Declare 3.40 as the minimum. Test on 3.44 as well. Use `qgis-plugin-ci` for packaging.	Adopt, add 3.44

IMPORTANT

Rule 6 in `AGENTS.md` says "do not introduce new cloud services without approved requirement". Colab is a new cloud compute service, so your approval of this plan counts as the approved requirement. I will record that in `stratumro-brain/inbox/` as a proposed decision. I will not edit `canon/` .

Layer mapping

Layer	Develop with	Runs on
L1 Geo Core	Antigravity (Gemini Flash) with Copilot	WS or LAPTOP. Colab only for bulk tiling if the laptop is too slow.
L2a Roof Segmenter	Claude for design, Gemini for scripts	**Train on Colab** (T4 or L4). **Infer on LAPTOP.**
L2b SAM 2.1	Gemini for scripts, Claude for evaluation	**Batch on Colab**, interactive on LAPTOP.
L3 Regularization	Claude and Copilot	WS or LAPTOP
L4 Triage	Gemini and Copilot	LOCAL, under 35 ms
L5 Verification	Claude (logic), Gemini Flash (routine)	CLOUD or WS fallback. Colab for VLM evaluation only.
L6 Human Gate and Feedback Loop	Claude for review	QGIS desktop. **Retraining on Colab.**

> NOTE
>
> An **AI Pro subscription is not the same as Gemini API access**. For L5's runtime "Gemini Flash Cloud API" you may need a separate API key. I can't confirm that from the docs I could read.

Proposed changes

A. Documentation (new, no code)

[NEW] `docs/HYBRID_DEV_WORKFLOW.md`

- The tool-role matrix and the layer mapping above.
- A **handoff protocol**: where each tool leaves state and what the next one reads.
- A **quota strategy**:
  - Claude Pro: planning and review.
  - Gemini Flash: bulk edits.
  - Copilot: completions.
  - Colab: check `colab usage` before every job.
- A **routing cheat sheet**: "if the task is X, open tool Y".

[NEW] `docs/architecture_v2_dev_plane.md`

- A Mermaid diagram that **references the same L1–L6 IDs** as v2 and adds the dev and compute plane beside it.
- A new `COLAB` tag next to the existing `WS`, `LAPTOP`, `CLOUD` and `LOCAL` tags.
- The original PNG is not touched. If you want, I can also render this as a PNG named `fig1b_dev_plane.png` next to the original.

B. Agent instructions (new files and append-only edits)

[MODIFY, append only] `AGENTS.md`

Add one short section at the end, "Hybrid toolchain", that links to `docs/HYBRID_DEV_WORKFLOW.md`. No existing line changes.

[MODIFY, append only] `CLAUDE.md`, `GEMINI.md`

One added line each. For Claude Code it is `@AGENTS.md` plus a pointer to the workflow doc.

[NEW] `.github/copilot-instructions.md`

Short, and it points to `AGENTS.md`. It states the golden rules in 6 lines: EPSG:3844, no invented coordinates, `ADMIS` is not an approval, and so on.

[NEW] `.agents/rules/hybrid-routing.md`

Has valid `trigger: model_decision` frontmatter. It tells agents when to propose a Colab job instead of running heavy work locally.

[NEW] `.agents/mcp_config.json`, `.mcp.json`, `.vscode/mcp.json`

These are mirrors of the existing `mcp_config.json` for Antigravity, Claude Code and Copilot. The root `mcp_config.json` is **left as is**. Drift between the mirrors would be a risk, so I'll add a small check script.

[NEW] `stratumro-brain/inbox/D-005-hybrid-colab-compute.md`

A proposed decision, following `MEMORY_RULES.md`. Promotion to `canon/` stays with you.

C. Colab compute scaffold (new, small, non-invasive)

[NEW] `compute/colab/README.md`

- WSL2 setup: `uv tool install google-colab-cli`, auth, and `colab usage`.

- Pin `google-colab-cli>=0.7.4` (0.7.3 was yanked).

[NEW] `compute/colab/colab_job.ps1` and `colab_job.sh`

A PowerShell wrapper that calls `wsl` and runs the shell script. The script runs: `new` → `upload` or Drive mount → `exec` → `download` → `stop`. It checks that `colab stop` ran, so compute units aren't burned.

[NEW] `compute/colab/jobs/*.yaml`

One job card per heavy task: `train_l2a`, `refine_sam21`, `vlm_eval`, `retrain_from_feedback`. Each declares:

- GPU type
- inputs and outputs
- the checkpoint path
- a time limit

[NEW] `compute/colab/scripts/train_l2a.py`

- `smp.UnetPlusPlus(in_channels=4)`.
- Normalise RGB with ImageNet statistics.
- Clip nDSM to 0–40 m, scale it and set nodata to 0.
- Geometric augmentation on all 4 channels. Photometric augmentation on RGB only.
- Dice + BCE loss.
- A **spatial split** using the diagram's numbers (Train 126, Val 22, Test 31, with a 30 m buffer).
- Resumable from the latest checkpoint.
- Writes `metrics.json`.

[NEW] `compute/colab/scripts/refine_sam21.py`

- `sam2.1_hiera_tiny`, bf16 autocast.
- One `set_image` per tile, then a batched `predict(box=...)` with the L2a masks or boxes as prompts.
- Logs peak VRAM.

Data contract (applies to every job)

- Inputs: 4-channel tiles in EPSG:3844, plus a `manifest.json` with SHA-256 hashes, CRS, resolution and the split.
- Outputs: `.pt` checkpoint, `metrics.json`, `run.log`, and the git SHA that produced them.
- Results come back as **proposals**. They never auto-approve, which keeps golden rule 3 intact.

- L2a stays recommended only if it **beats the SAM baseline on the frozen test area** (decision D-001).

> [!CAUTION]
> Orthophoto and cadastral tiles would be uploaded to **your private Google Drive and a Colab VM**. If any of that data has a licence or confidentiality limit (USAMV, ANCPI), tell me and I'll limit Colab to the public or cleared subset.

D. Notebook housekeeping (optional)

[MODIFY] `StratumRO_Pipeline.ipynb`

Replace the hard-coded `C:\Users\lefpa\Desktop\...` path with a config value, **only if you approve**. It is not part of the core plan.

Safety rules for execution

- Work on a **new branch** from the current one. I won't touch the 11 untracked `perplexity AI/` files or the modified `graph.json`.
- Run `git status -s` first, as `AGENTS.md` requires.
- No push, merge or history rewrite.
- No secrets in any file. `.env` is not read or copied.
- No GPU job is started without you telling me.

Open questions

> [!IMPORTANT]
> 1. **WSL2:** is WSL2 installed, or should I include setting it up?
> 2. **Claude access:** do you use Claude Code (terminal or VS Code extension) in this repo, or can you pick Claude inside Antigravity on your plan? This decides how I write the Claude instruction file.
> 3. **Colab data approval:** can orthophoto and survey tiles go to your private Drive and Colab (see the CAUTION above)?
> 4. **Gemini API key:** do you have a separate Gemini API key for L5 runtime calls, or should L5 default to local Qwen-VL?
> 5. **QGIS 3.44 test lane:** may I add a 3.44 job to CI later? It is not part of this plan.

Verification plan

Automated

```
powershell
git status -s
python scripts/validate_project.py --mode fast
python -m unittest discover stratum_ro/test
python stratumro-brain/scripts/lint_vault.py
python scripts/check_mcp_mirrors.py      # new: the three MCP mirrors match the root file
```

- A markdown link check on the new docs.
- Frontmatter check on `.agents/rules/*.md`, since invalid files are silently dropped.

Manual

1. In WSL: `colab --version` and `colab usage`.
2. A smoke job on a **T4** with a tiny synthetic 4-channel dataset. It should run for a few minutes, write a checkpoint, download it and call `colab stop`.
3. Open the repo in the Antigravity IDE, Claude Code and VS Code with Copilot. Confirm each one reads the shared rules.
4. Confirm that `diff` against the original `AGENTS.md`, `CLAUDE.md` and `GEMINI.md` shows only added lines.
5. Confirm that no existing file under `docs/`, `.agents/` or `stratumro-brain/canon/` was changed.

Deliverable after approval

A `walkthrough.md` summarising what was added, what ran, and the results of the smoke test.